EECS 245, Fall 2026

LEC 15 Inverses; Projecting onto Column Space

→ Read: Ch. 6.2-6.4

Agenda

- Recap: what makes a matrix invertible?
  - 2x2 closed form
  - Finding A^{-1} in practice
  - The determinant
- Projecting onto $\text{colsp}(X)$
  - In other words, linear regression!

Announcements

- HW 6 due Sunday
- HW 5 scores up + Grade Report updated
- Make sure to watch the YouTube playlists for the last few lectures!

Suppose A is an invertible matrix.

What does that mean?

- A is square $n \times n$
- A is "full rank": $\text{rank}(A) = n$
  - also, A's cols are linearly independent
  - also, A's rows are linearly independent
  - also, $\text{nullsp}(A) = \{\vec{0}\}$
  - also, $\det(A) \neq 0$

} all equivalent for square matrices!

Suppose A^{-1} exists.

What properties does it satisfy?

→ A^{-1} is the unique matrix such that

$$AA^{-1} = A^{-1}A = I$$

Remember:

$$T(\vec{x}) = A\vec{x}$$

$$\vec{x} \longrightarrow \boxed{\text{multiply by } A} \longrightarrow A\vec{x}$$

How do you actually find an inverse?

$$A = \begin{bmatrix} 1 & 5 \\ 2 & 12 \end{bmatrix} \qquad A^{-1} = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & 5 \\ 2 & 12 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x & y \\ z & w \end{bmatrix}}_{A^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{I}$$

$$\underbrace{\begin{bmatrix} 1 & 5 \\ 2 & 12 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x & y \\ z & w \end{bmatrix}}_{A^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{I}$$

$x + 5z = 1$ $\qquad$ $y + 5w = 0$

$2x + 12z = 0$ $\qquad$ $2y + 12w = 1$

$$\begin{bmatrix} 1 & 5 \\ 2 & 12 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

2×2 inverse formula

remember this!

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

determinant, det(A)

$$A = \begin{bmatrix} 1 & 5 \\ 2 & 12 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 12 & -5 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & -5/2 \\ -1 & 1/2 \end{bmatrix}$$

A^{-1}

Aside: determinant

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(A) = ad - bc$$

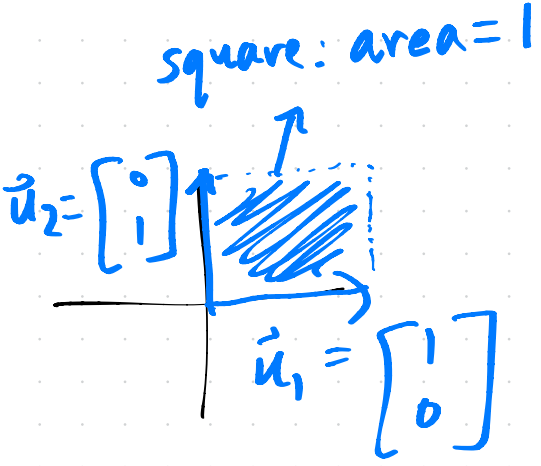
scalar

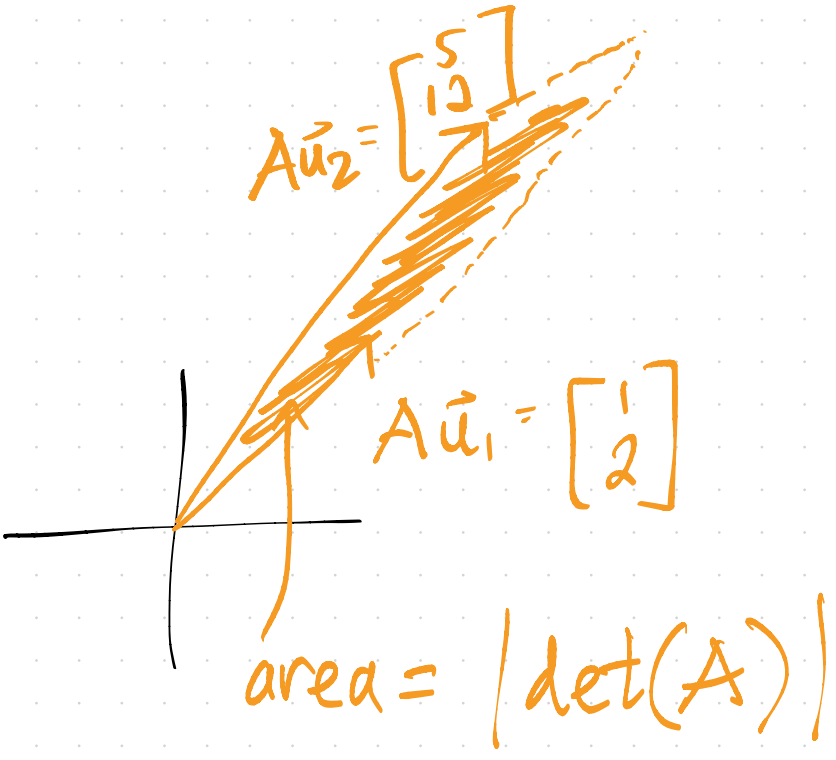
e.g. $A = \begin{bmatrix} 1 & 2 \\ 6 & 12 \end{bmatrix}$ $\det(A) = 0$

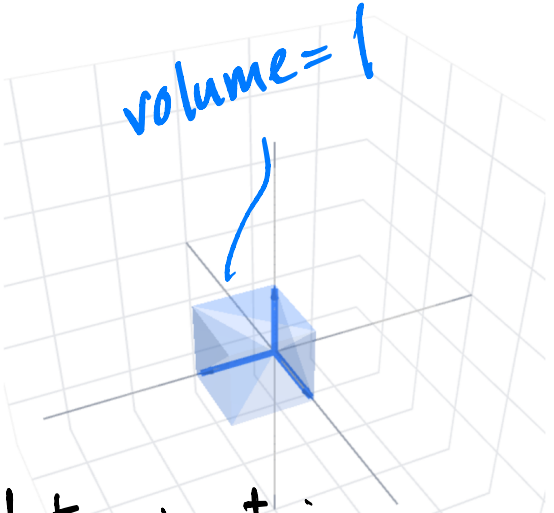
A is not invertible!

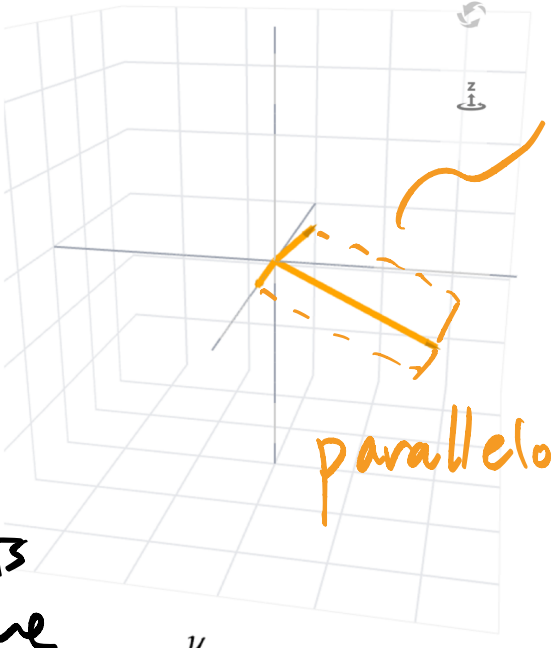
What does the determinant tell you?

$$A = \begin{bmatrix} 1 & 5 \\ 2 & 12 \end{bmatrix}$$







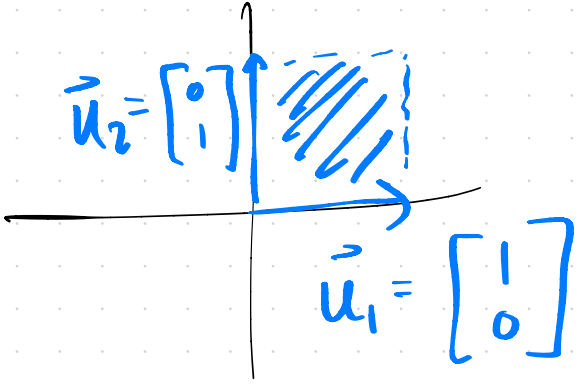


determinant:

= n-dimensional volume of object that results from transforming the n standard basis vectors

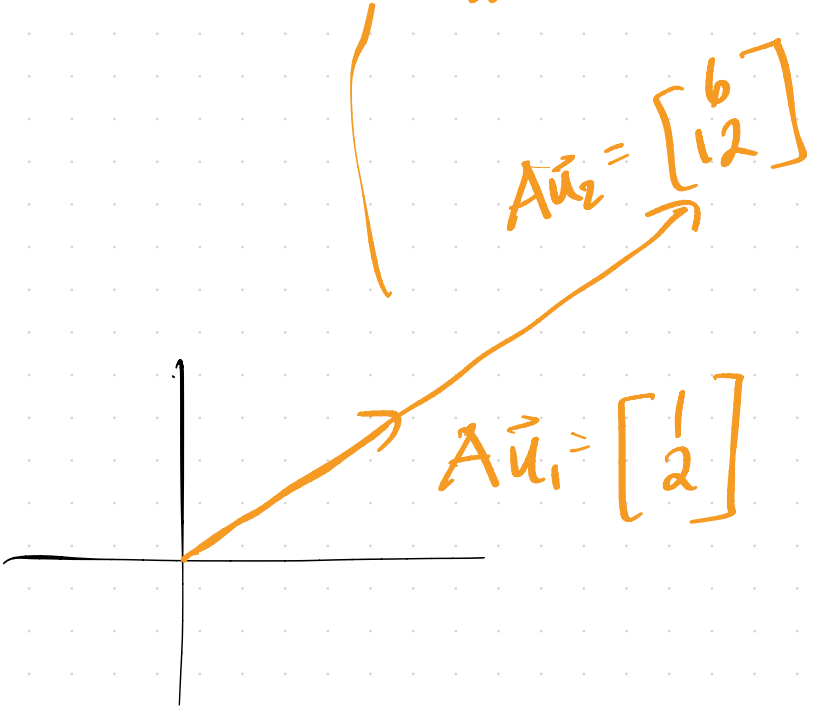
What if $\det(A) = 0$?

$$A = \begin{bmatrix} 1 & 6 \\ 2 & 12 \end{bmatrix}$$



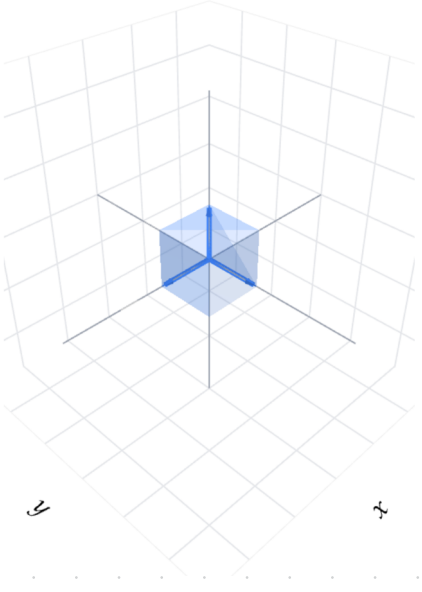
no enclosed parallelogram!

$\det(A) = 0$

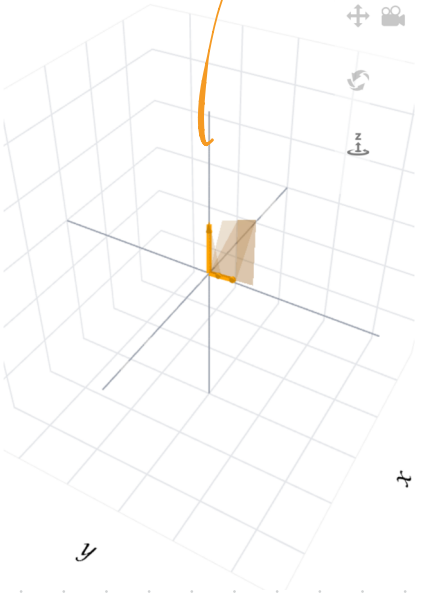


$$L = \begin{bmatrix} 1 & 1/2 & 0 \\ 1 & 1/2 & 0 \\ 1 & 1/2 & 1 \end{bmatrix}$$

$\vec{u}_x$, $\vec{u}_y$, and $\vec{u}_z$



$L\vec{u}_x$, $L\vec{u}_y$, and $L\vec{u}_z$



Overview
Linear Transformat
Activity 1
Finding the matri: transformation
Different domain
Lines stay lines
Example: comput
Linear Transformat
Scaling
Rotations and Ort
Composing Trans
Shears
Non-Invertible Tra
Arbitrary Matrices
The Determinant
Activity 2

$\det(L) = 0$, because enclosed shape has no volume!

Properties of the determinant

$\det(A) = \det(A^T)$

- swap 2 rows, det multiplied by -1
- swap 2 cols, det multiplied by -1
- multiply a row/col by c, det multiplied by c
- $\det(I) = 1$

determinant uniquely defined by these properties

There exists a 3x3 determinant formula

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

then $\det(A)$ is

$$\det(A) = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

determinant symbol

Finding inverses in general

$$A_{n\times n}\vec{x} = \vec{b}$$

→ n equations, n unknowns

If A^{-1} exists, the unique $\vec{x}$ that satisfies $A\vec{x} = \vec{b}$

$$(A^{-1}A)\vec{x} = A^{-1}\vec{b}$$

$$\vec{x} = A^{-1}\vec{b}$$

Issue: finding A^{-1} is hard!

→ $A\vec{x} = \vec{b}$ n equations, n unknowns;

→ A^{-1} : n^2 unknowns

→ better to solve directly

Suppose A, B $n \times n$ matrices,

A invertible,

B is not,

is AB invertible?

no!

If AB is invertible, we must be able to undo

$$T(\vec{x}) = AB\vec{x},$$

i.e. for every $\vec{y} \in \mathbb{R}^n$, must be a unique soln

$$\vec{y} = AB\vec{x}$$

$$A^{-1}\vec{y} = B\vec{x}$$

but B is not invertible!

- If A, B both $n \times n$ invertible,

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$\underbrace{ABB^{-1}}_{I}A^{-1} = AIA^{-1} = I$$

- $\det(AB) = \det(A)\det(B)$

only applies to square A, B

Suppose AB is invertible

but you know nothing else about A or B.

Are A and B invertible?

→ not necessarily, because A, B don't have to be square!

→ if they are square, they must be invertible

Remember, if A is $n \times d$,

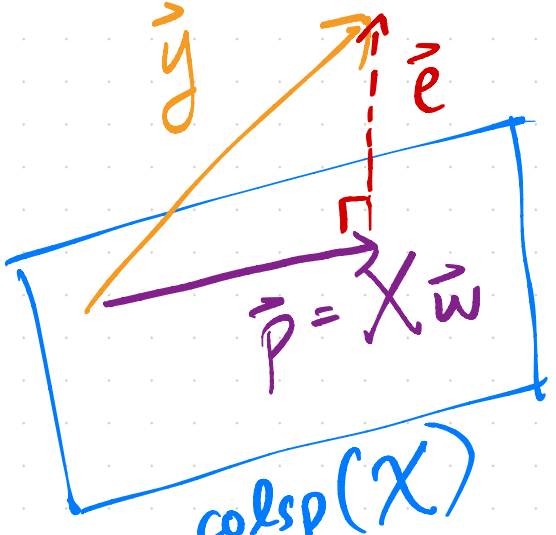
A^TA $d \times d$ → interpretation: dot products of pairs of cols

$$A^TA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$ → invertible!

$$A = \begin{bmatrix} 3/5 & 4/5 \\ 0 & 0 \\ 0 & 0 \\ 4/5 & -3/5 \\ 0 & 0 \end{bmatrix}$$ → not square, so definitely not invertible

Suppose X $n \times d$

$\vec{y} \in \mathbb{R}^n$



↑

subspace of $\mathbb{R}^n$